



THE UNIVERSITY OF HARIPUR

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WEEK 5 INTERNSHIP PROGRESS REPORT

Remote Internship at FlyRank AI

STUDENT INFORMATION

University Name	University of Haripur
Student Name	Muhammad Abdullah Awan
Roll Number	F23-0332
Department	Department of Information Technology
Program	BS Artificial Intelligence
Semester	7th Semester
Company	FlyRank AI
Internship Type	Remote Internship
Internship Duration	July 2026 – August 2026 (8 Weeks)
Report Period	Week 5 (July 29 – August 4, 2026)

Submitted To:

Dr. Dawar Khan

Submitted By:

Muhammad Abdullah Awan

1. Introduction

FlyRank AI is an AI-powered organic growth platform that helps businesses improve their visibility across traditional search engines and AI answer engines such as ChatGPT, Perplexity, and Gemini. The platform automates large parts of the content lifecycle - research, scoring, and optimization - primarily serving D2C, SaaS, and agency clients who need scalable organic growth.

This report covers Week 5 of my remote Machine Learning Engineering internship at FlyRank AI, under the Applied Search Intelligence track. It summarizes the assignment completed, the reasoning behind it, and the skills developed.

2. Objective of Week 5

Objective: To build the first machine learning model for my lane and compare it honestly against the Week 4 rule-based baseline, on the exact same data split and metric, before believing any score.

3. Week 5: Model Building & Baseline Comparison

Task Requirements: Choose a modeling method that fits the ranking/scoring question; design a valid, leakage-safe train/test split; train the model(s) and produce a single comparison table (baseline vs. model, same split, same metric, plus the base rate); interpret feature importances and read concrete wrong predictions rather than trusting the score alone.

Research Performed: Reviewed this week's session material on matching a modeling method to the shape of the question (ranking needs scores evaluated at precision@K, not plain classification accuracy), and read the assigned research paper to prepare for next week's methodology review. Compared Logistic Regression, Decision Tree, and Random Forest as candidate methods for the lane.

Summary of Work Completed: Trained a Logistic Regression model (fully readable, one coefficient per feature) and a Random Forest model (stronger, still explainable via feature importance) to rank pages by decline risk. Used a client-holdout split (grouped by client_id, matching the Week 4 baseline's split design) to prevent client-level leakage between train and test, and confirmed zero client overlap. Recomputed the Week 4 baseline rule on the identical test split for a fair comparison, and built one final table showing base rate, baseline rule, Logistic Regression, and Random Forest side by side on Precision@50 and ROC-AUC. Random Forest gave the strongest lift over the baseline. Reviewed the model's feature importances - the top features matched the signals already verified in Week 4 (staleness and visibility), ruling out an obvious leakage explanation for the improvement - and manually inspected three concrete wrong predictions from the top 50 ranked pages to understand where and why the model fails.

What Was Learned: Learned that a model's score means nothing without a same-split, same-metric baseline comparison sitting next to it, and that reading concrete wrong cases - not just the headline metric - is what separates a trustworthy result from a decorative one. Also confirmed that a model's top features should make sense given what was already verified about the data; a 'too good to be true' top feature is a leakage warning sign, not a win.

4. Skills Developed

Week 5 contributed to the development of the following technical, analytical, and professional skills:

Skill	Description
Model Selection	Matched the modeling method (ranking via classifier probability) to the actual shape of the business question.
Grouped Validation	Designed and verified a client-holdout split to prevent leakage between train and test.
Honest Benchmarking	Built a single baseline-vs-model comparison table on identical data and metrics, rather than reporting model results in isolation.
Error Analysis	Interpreted feature importances and manually reviewed concrete wrong predictions to explain model failure modes.

5. Challenges Faced

- › Keeping the comparison fair: Ensuring the Week 4 baseline rule was recomputed on the exact same client-holdout test split - not just reused from the earlier notebook - required care to avoid an apples-to-oranges comparison.
- › Resisting the urge to over-trust the score: Interpreting feature importances and manually checking wrong cases, rather than stopping at the headline Precision@50 number, took deliberate extra effort.

6. Knowledge and Experience Gained

Week 5 provided hands-on experience in the full honest-modeling workflow: choosing a method that fits the question, validating with a leakage-safe split, building a single fair comparison table, and reading errors before trusting a score. This is the first true modeling milestone of the capstone project.

7. Conclusion

Week 5 moved from a transparent rule-based baseline to a first machine learning model, evaluated honestly against that same baseline on identical data and metrics. The Random Forest model showed a real, explainable improvement over the Week 4 baseline, with feature importances that align with previously verified signals rather than suggesting leakage.

Looking ahead, the objective for the next reporting period is to inspect a research paper's methodology with an engineering lens and continue refining model validation practices.